

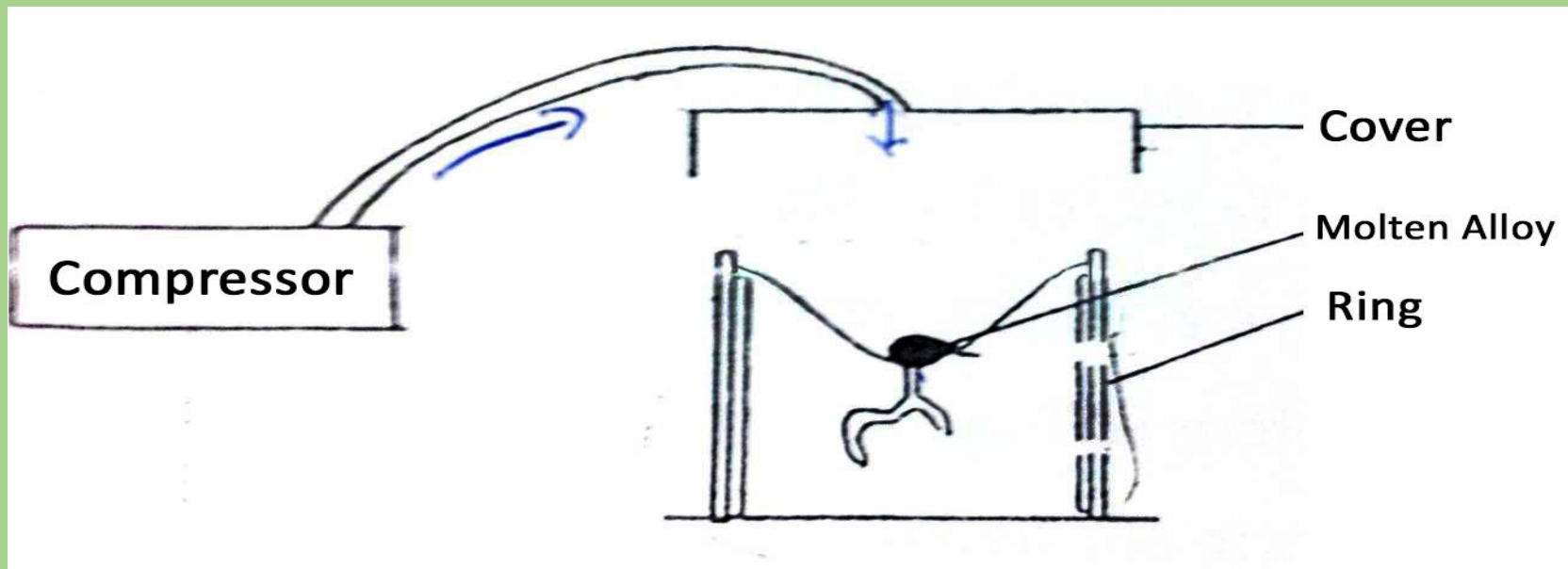
Casting machines

By

Dr Sherif Refat

Casting

- A) **Air Pressure casting machine**
- The precious and semi precious alloys are melted over the ring at the top of the crucible using a torch to melt the alloy.
- Compressed air is used to force the molten alloy into the mold.



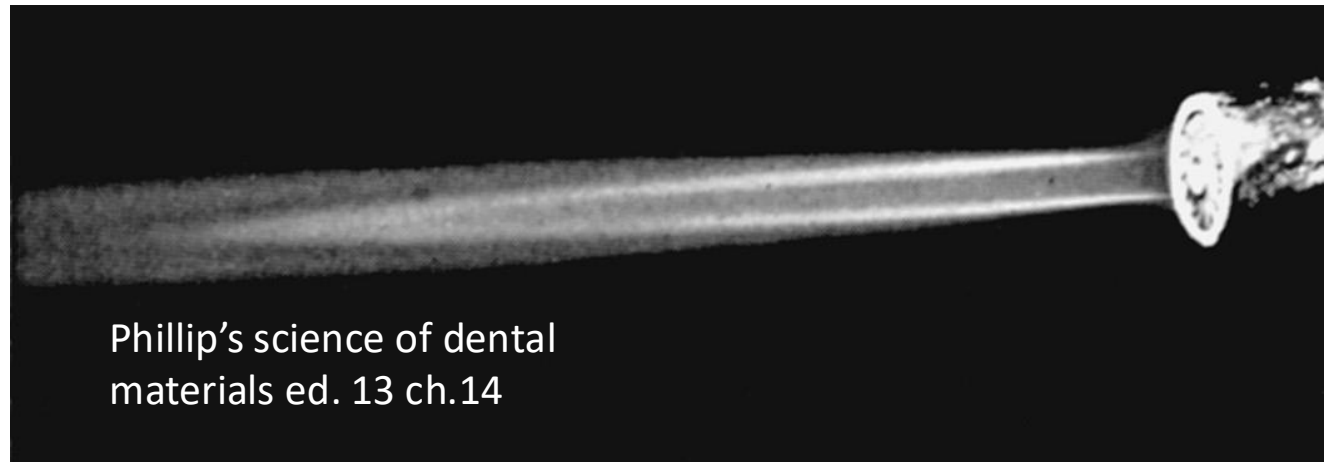
- **B-Centrifugal casting machines:**
- Most casting of alloys is made in a **centrifugal casting machine**.
- **Centrifugal casting machines** operate on the physical principle of **centrifugal force**, in which the molten material is accelerated outward by rapid spinning.
- The casting machine is driven with a spring or electric motor.

- **FLUXES:**

- The casting alloy is heated in a crucible made of a heat-resistant ceramic.
- A flux is usually added to minimize the formation of oxides, *because oxides hinder the heating and casting of the alloy and decrease the final quality of the casting.*
- The type of flux :used depends on the temperature of casting, the type of heat source, the type of casting alloy, and the type of investment.
- **may** be in case of **some precious alloys** made **from equal parts of fused borax powder ground with boric acid powder**. The boric acid aids in retaining the borax on the surface of the alloy.

- **When the alloy is completely molten**, the casting ring is removed from the burn out oven and is placed behind the crucible and the crucible–ring assembly is spun rapidly.
- Spinning accelerates the alloy into the casting ring and into the space previously occupied by the sprue and wax pattern.
- Casting alloys may be heated by blowtorch or electric current.
- ***The most common method for heating low-temperature alloys (<1200° C) is by blowtorch***
- ***An electric current is used for higher temperatures alloys, using a process called induction casting***

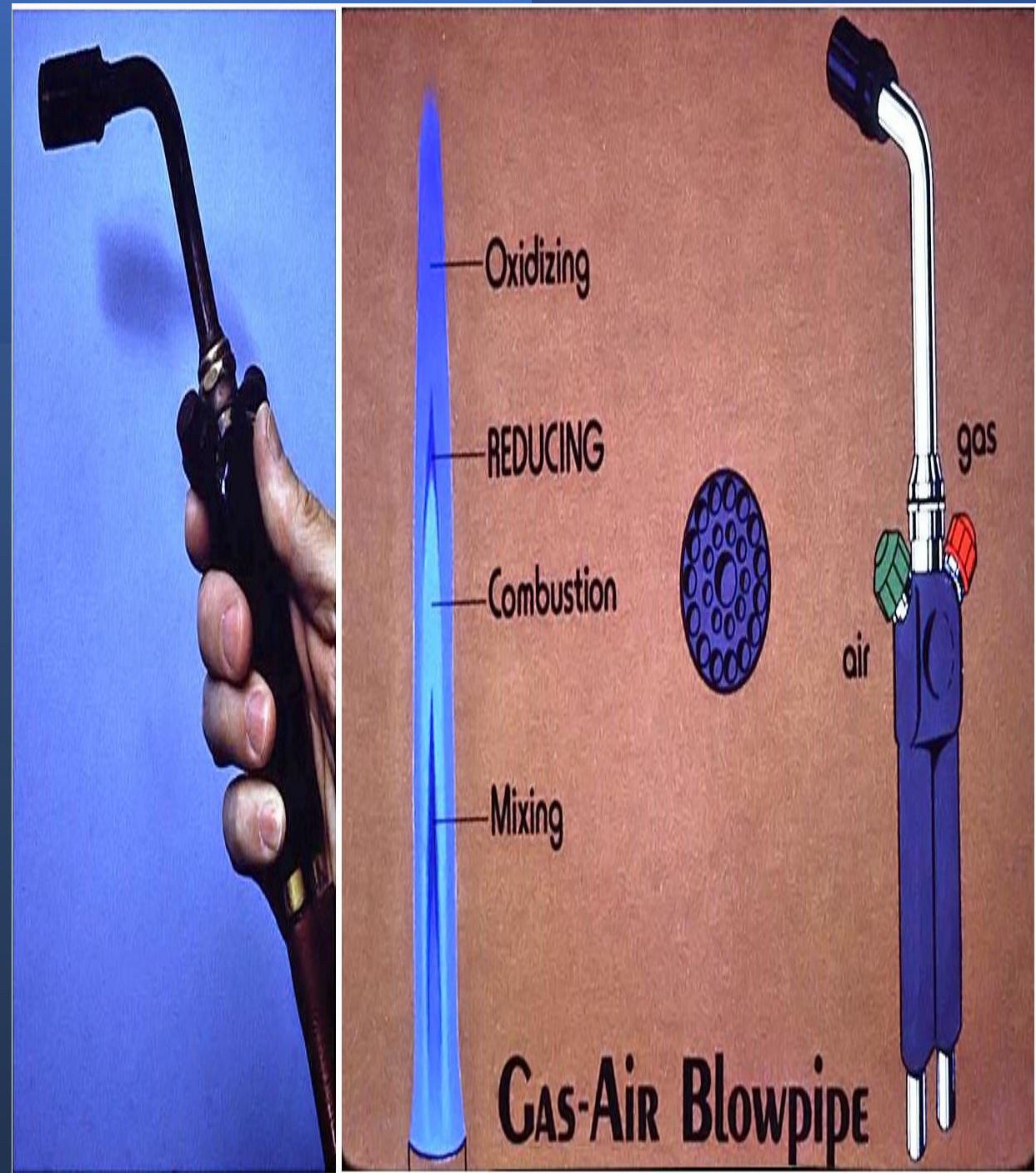
- the proper heating of the alloy is critical to a successful casting:
- *If heated too little*, the alloy will be too viscous to flow into all of the finest details of the casting ring mold.
- *If heated too much*, the alloy may be damaged by the oxidation of elements, or the investment may crack from the thermal shock
- The torch flame is generated from a gas mixture of propane and air, natural gas and air, acetylene and air, or acetylene and oxygen.



Parts of the blowtorch flame:

1) **Mixing Zone** – First zone coming out from the nozzle . Mixing of air & gas takes place here before combustion. No Color . No heat .

2) **Combustion Zone** – Second zone covering the first one. Partial Combustion of air & gas takes place here. Green in Color . Zone is oxidizing & should always be kept away from molten alloy during fusion.



3) Reducing Zone – Third zone located beyond the tip of the combustion zone. Light blue in Color. It is the hottest part of the flame . This area should be kept constantly on the metal during melting.

4) Oxidizing Zone– Outer zone in which complete combustion occurs with oxygen & air. Not used to melt the alloy. Its temperature is lower than the reducing zone & also oxidizes the alloy.



Cooling:

After casting, cooling is either by allowing casting ring to cool slowly on top of a bench or is submerged immediately in cold water.

For some gold-based alloys, bench cooling produces a harder, stronger state of the alloy, and rapid cooling in water (also called quenching) produces a softer condition.

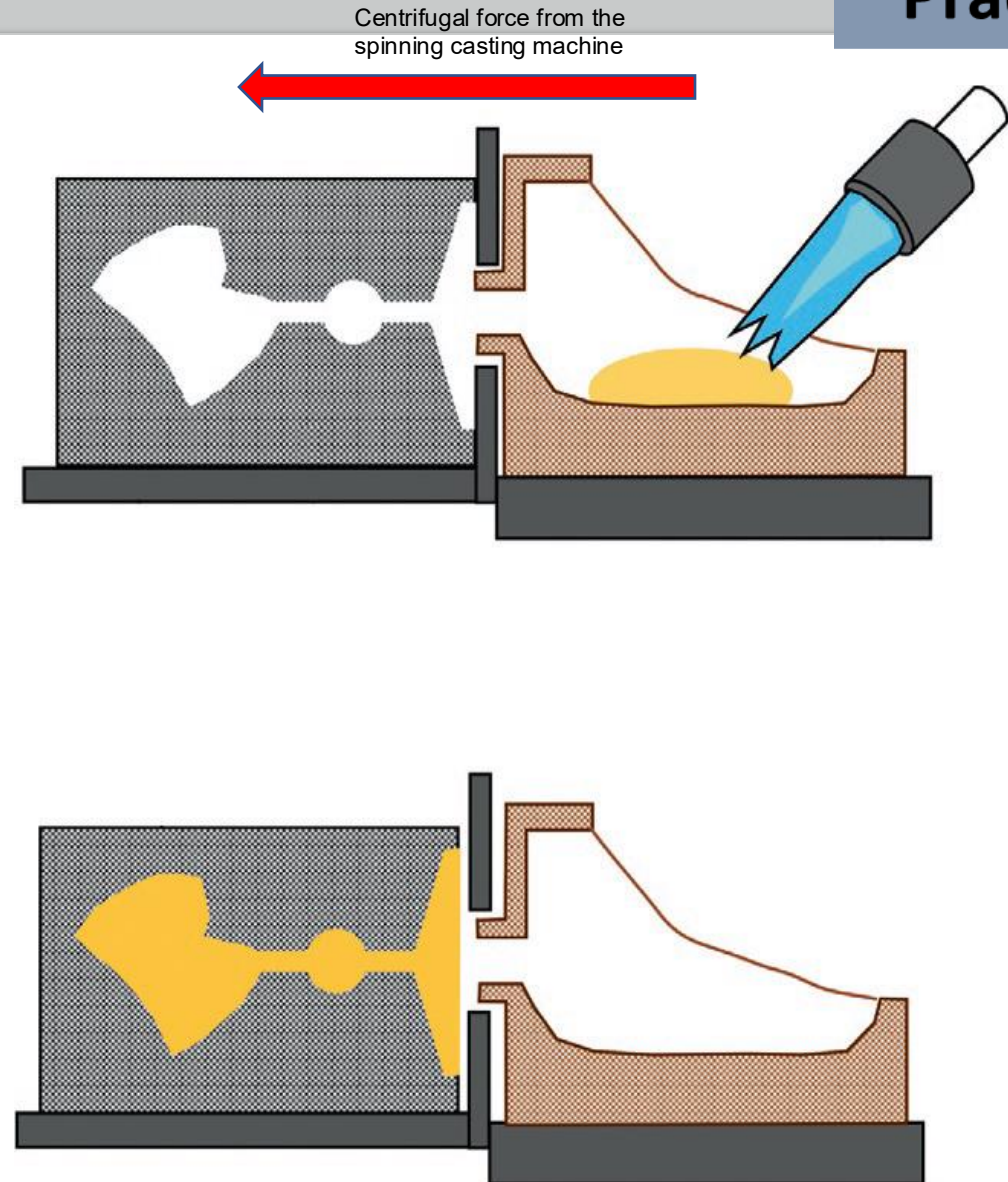
However, many types of alloys do not obey this principle.

You choose type of cooling according to the desired mechanical properties

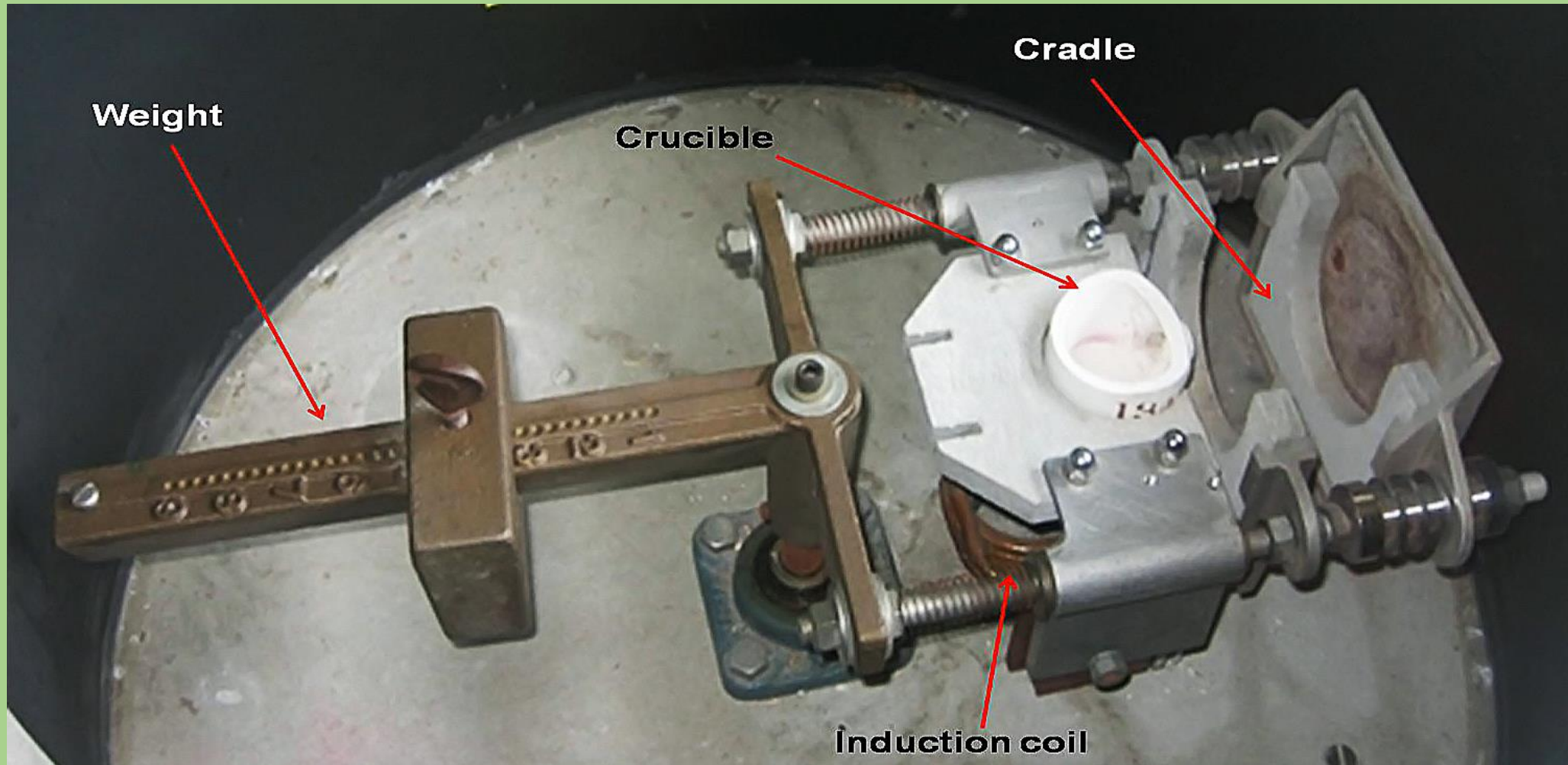
FIG 12-6 Schematic diagram of the alloy casting process.

After burnout of the wax from the ring, the ring is moved from the burnout oven to the cradle in the casting machine (left). At this point, the alloy has been melted with the torch in the crucible (right), which remains in place to maintain the molten state. As soon as possible, the casting arm is released and the spinning force of the casting arm generates centrifugal force (red arrow above) that drives the molten alloy toward the ring.

(Dental materials foundations and applications 11 ed)

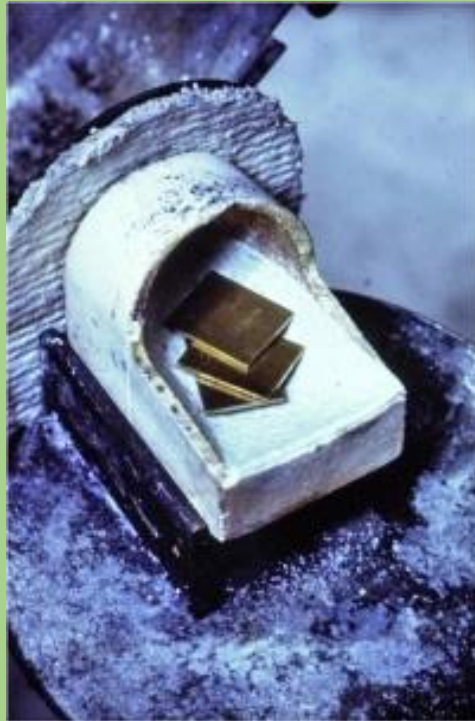


Practical



<https://mydentaltechnologynotes.wordpress.com/2018/08/12/casting-procedures-in-dentistry/>

Practical



Gold alloys placed in the crucible



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Practical

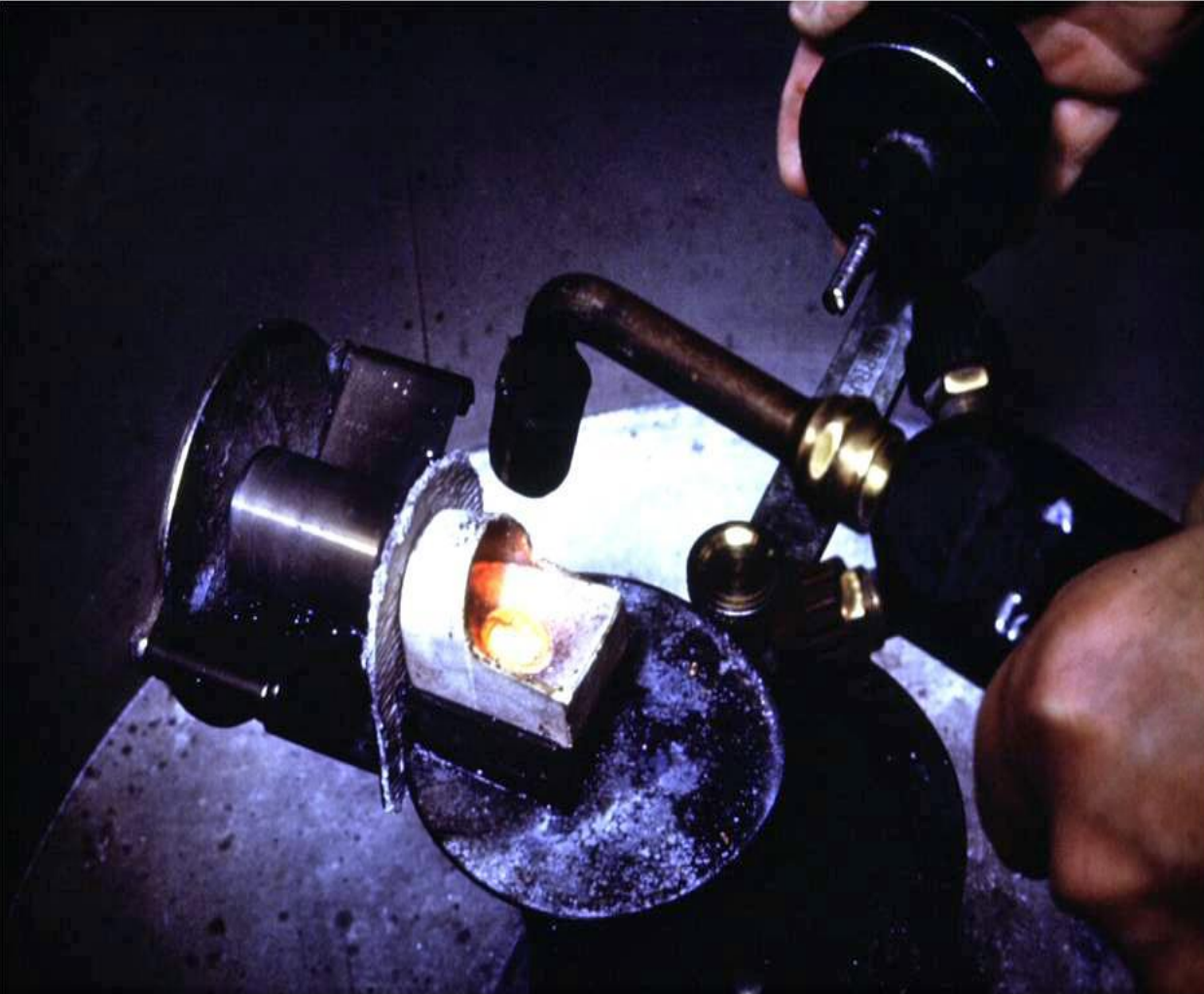
Melt gold with reducing flame



Shiny ball like appearance of gold alloy

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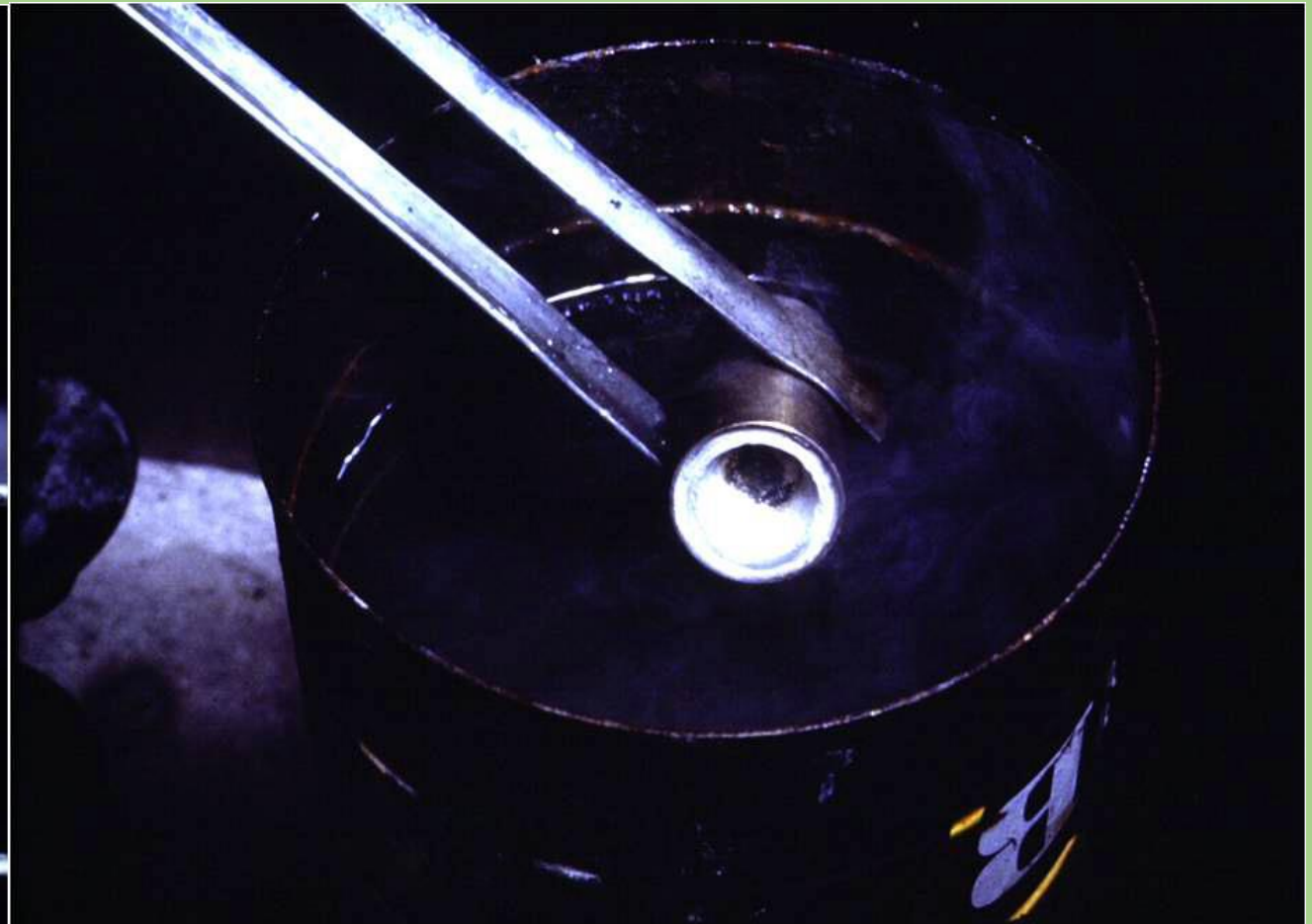


Releasing the lock



Casting takes place

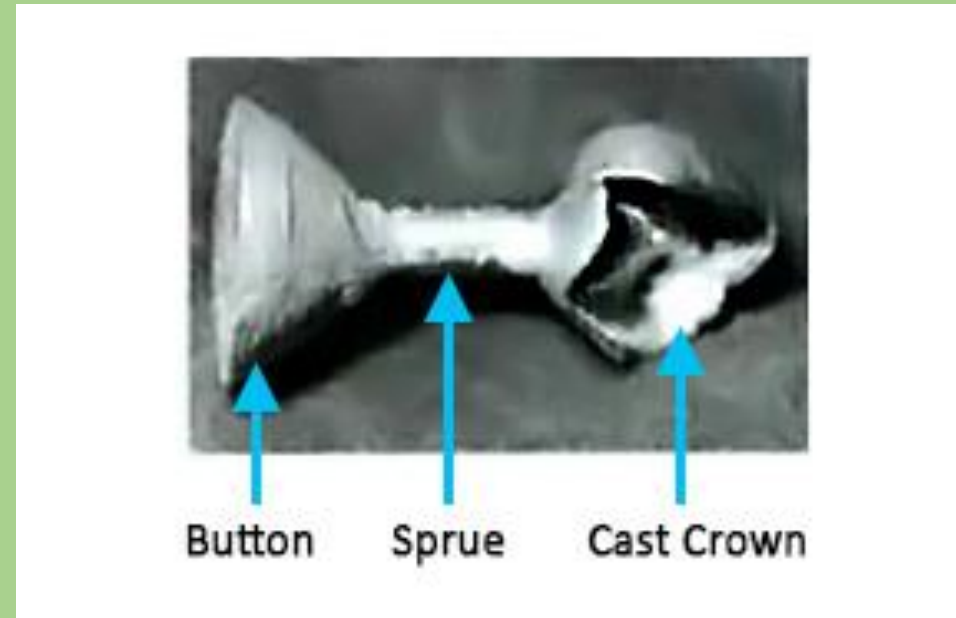
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Quench ring in cold water

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- **C) Vapor Pressure casting machine**

- The precious and semi precious alloys are melted over the ring at the top of the crucible using a torch to melt the alloy.
- A metal cover lined with wet liner is adapted over the ring . The produced Vapor is used to force the molten alloy into the mold.

- **D) Vacuum casting machine**

- This machine evacuates the mold before it is filled with molten alloy.
- Vacuum improves the mold cavity filling.

Riyadh Colleges Of Dentistry and Pharmacy

Department of Prosthodontics



Educational Video Series



References

- 1- Dental applications and foundations 11th ed.
- 2-
<https://mydentaltechnologynotes.wordpress.com/2018/08/12/casting-procedures-in-dentistry/>
- 3- Philips 's science of dental materials 13ed. CH.14

The background is a complex, abstract geometric pattern composed of numerous 3D cubes and 2D hexagons. These shapes are arranged in a way that creates a sense of depth and movement. The color palette is diverse, featuring shades of purple, teal, dark blue, light green, and grey. The shapes are scattered across the frame, with some appearing more prominent than others, creating a dynamic and modern aesthetic.

Thank you